Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “work” to compare your model or result. Sentence starters are optional.

trial: one run or repetition of a test

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A 6 N force acts at 120° to a 5 m displacement.
Calculate its work and explain the sign.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and record actual trials separately.

Worked reasoning: Supplied measurements give net work 12 J and kinetic-energy change 11 J. Their 1 J difference could reflect uncertainty or omitted work; it is evidence to investigate.

Your turn: Draw two bars for these measurements and label one check of the experiment.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

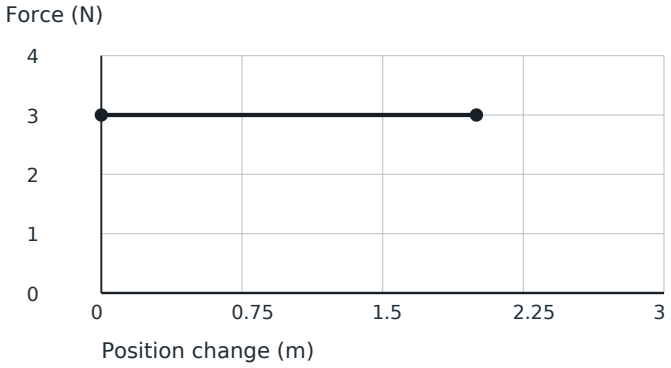
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

The supplied force-displacement graph stays at 3 N over 2 m.
Calculate work.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A 0.50 kg lab cart speeds up from 0.40 to 1.20 m/s. Measured pulling work is +0.35 J. Complete the energy account.
Find ΔK and infer the work of resistance if every other force does zero work.

Word help: infer = use evidence to reach a conclusion

Sentence starter: I found … [unit]. This value means … in this situation.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Create a trial table with mass, initial/final speed, work and ΔK.
Name two causes of disagreement that deserve checking.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg projectile at 60 m/s sticks to a stationary 3.6 kg cart.
Find common speed and the fraction of initial kinetic energy remaining.
Use before and after snapshots to organize the momentum account.

Word help: initial = at the beginning of the interval; subscript i

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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Continue down each column.

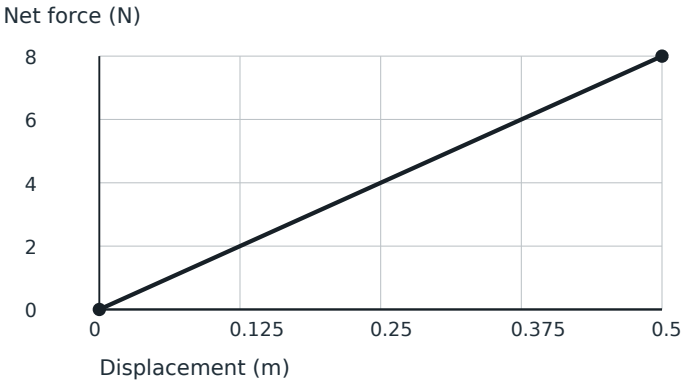
Sentence starters are optional. You may plan in any language.

07 Exit check

A force rises linearly from 0 to 8 N over 0.50 m.
Find work. If a 1 kg cart begins at rest and this is net work, find final speed.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart speeds from 1 to 3 m/s while measured pulling work is 10 J. Infer friction work and mean friction magnitude over 2 m if other work is zero.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: infer = use evidence to reach a conclusion

Sentence starter: Both … . They differ in … because … .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3